Separation Logic I

The heap, the separating conjunction and the frame rule

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Lecture 10

COMSM0067 — Advanced Topics in Programming Languages
University of Bristol

Where we are

Two lectures of Hoare logic. It reasons about variables.

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Today we add the heap — and that assumption stops holding.

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Add pointers to the IMP language: $[x]$ is "the heap cell that x points to".

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before: $[y] = 3$ ✓

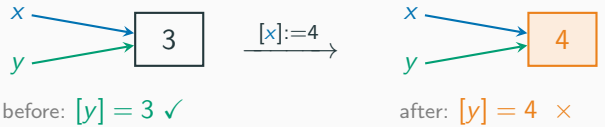
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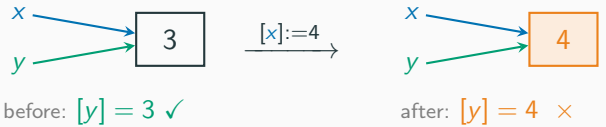
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The problem: aliasing

x and y hold the *same* address, so $[x] := 4$ also changes $[y]$. Substitution is syntactic: it cannot see that two *different names* denote *one cell*.

The model

States, now with a heap

A state is now two components:

$$\text{store } s : \mathrm{Var} \to \mathrm{Val} \qquad\qquad \text{heap } h : \mathrm{Loc} \rightharpoonup \mathrm{Val}$$

- The store is the old σ: variables to values. Total.
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We write $s, h \vDash P$. Assertions will now describe the shape of the heap, not just the values of variables.

Heap assertions

The empty heap

$s, h \vDash \mathsf{emp} \quad \text{iff} \quad \mathrm{dom}(h) = \varnothing.$

Points-to: a single cell

$s, h \vDash x \mapsto v \quad \text{iff} \quad \mathrm{dom}(h) = \{\, s(x) \,\} \quad \text{and} \quad h(s(x)) = s(v).$

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Read $x \mapsto v$ as: "the heap is exactly one cell, at address x, holding v".

Small axioms of separation logic

Commands for the heap

Four new commands, each with a small axiom — stated on the smallest heap that makes sense, its *footprint*.

mutation	$\{x \mapsto _\}\ \ [x] := a\ \ \{x \mapsto a\}$	
lookup	$\{x \mapsto v\}\ \ y := [x]\ \ \{y = v \wedge x \mapsto v\}$	(y fresh)
allocation	$\{\mathsf{emp}\}\ \ x := \mathsf{new}(a)\ \ \{x \mapsto a\}$	
disposal	$\{x \mapsto _\}\ \ \mathsf{dispose}(x)\ \ \{\mathsf{emp}\}$	

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We need a way to say "these cells are separate":

$$\{\, x \mapsto 3 \land y \mapsto 3 \land z \mapsto 7 \land x \neq y \land x \neq z \,\} \;\; [x] := 4 \;\; \{\, x \mapsto 4 \land y \mapsto 3 \land z \mapsto 7 \,\}$$

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The goal: local reasoning

We want to reason about the cells a command *touches*, and ignore the rest of the heap — without proving, every time, that "the rest" really is untouched.

The separating conjunction of separation logic

The separating conjunction

$P * Q$ — "P and Q, on disjoint heaps"

$s, h \vDash P * Q$ iff h splits as $h = h_1 \uplus h_2$ with

$$h_1 \perp h_2, \qquad s, h_1 \vDash P, \qquad s, h_2 \vDash Q.$$

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$P * Q$ asserts P and Q hold on separate parts of the heap. The split is an *existential*: "there is some way to divide the heap so that. . . ".

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This is the heart of separation logic.

The frame rule at work

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R might be a million other cells. We never re-verify them.

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Let's drop it.

Take $\{\mathsf{emp}\}\ \ x := \mathsf{new}(4)\ \ \{x \mapsto 4\}$, where $\mathrm{mod}(c) = \{x\}$.

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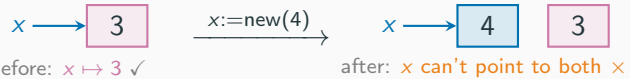
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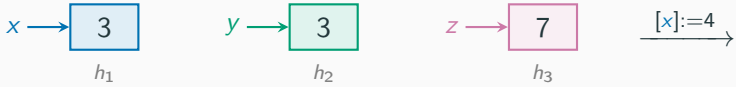
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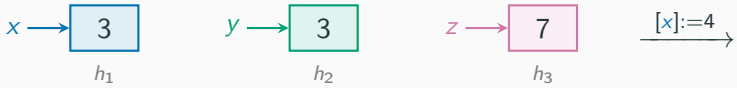
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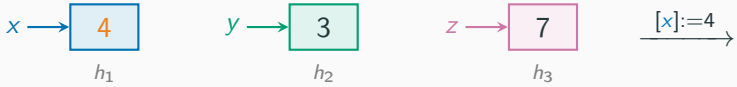


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Premise: the small axiom, on h_1 alone. Side condition: $[x] := 4$ modifies no variable in R✓

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Example: swapping two cells



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$$\{x \mapsto a * y \mapsto b\}$$
$$t := [x]; \quad u := [y]; \quad [x] := u; \quad [y] := t$$
$$\{x \mapsto b * y \mapsto a\}$$

Swap: the proof

Each step: a small axiom on one cell, framed with the other.

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$[x] := u;$ — mutation, frame $y \mapsto b$

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$[x] := u;$	mutation, frame $y \mapsto b$
$\{t = a \wedge u = b \wedge (x \mapsto b * y \mapsto b)\}$	
$[y] := t$	mutation, frame $x \mapsto b$
$\{t = a \wedge u = b \wedge (x \mapsto b * y \mapsto a)\}$	
$\{x \mapsto b * y \mapsto a\}$	consequence: drop $t = a,\ u = b$

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$\{x \mapsto b * y \mapsto a\}$ — consequence: drop $t = a,\ u = b$

Side conditions ✓: lookups modify only t, u; mutations modify nothing.

Wrap

Today

- State = store + partial heap. Assertions describe heap shape: emp, $x \mapsto v$.
- The separating conjunction $P * Q$ splits the heap into disjoint parts. $* \neq \wedge$: $*$ *builds in disjointness*.
- Small axioms on footprints + the frame rule = local, modular reasoning.

Next time

We can describe a single cell. Real programs use data structures (lists, trees) which are unboundedly many cells.

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Separation Logic II (Lecture 11): inductive predicates.

Questions?

ATiPL 2026 Minute Sheet